

B.Tech Major Project Presentation on **Support BOT for Automating L1 Triage**

Presented by

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ORGANIZATION OF PRESENTATION

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Introduction

- Development Operations is an important link in the chain of processes which a software needs to go through before being deployed or be commercially released.
- One of the main functions of the DevOps team is to respond to the requests raised or the incidents reported by the other teams involved in the development or testing or deployment of the software.
- These requests or concerns are raised as tickets through a portal named Service Now.

Motivation

- The motivation for this project is saving human work hours.
- The task of handling the tickets raised on the SNOW portal is not only tedious and monotonous but also involves a poor utilization of the technical skills of an individual.
- By automating the process, an individual`s technical acumen can be utilized in a more meaningful manner
- Also, the ticket handling process can be speeded up which will help in improving the performance of the software production chain.

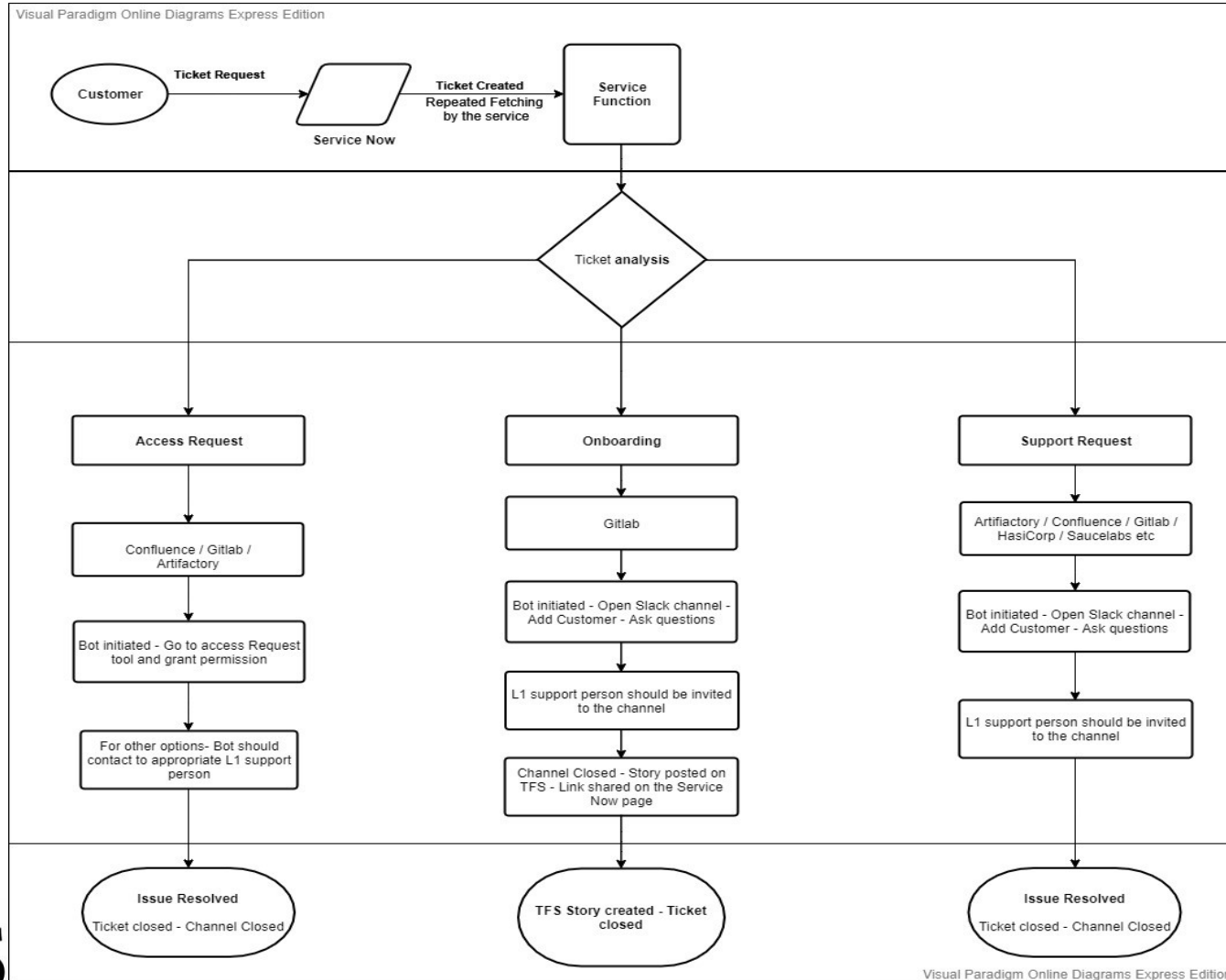
Objective

- To quickly fetch and resolve the tickets.
- To segregate the tickets based on priorities and kind of assistance required
- To ensure minimal manual intervention in the resolution of the ticket.
- Diverting the issue to a support person if the problem involved is beyond the bot`s ambit.

Literature Survey

Sl.no	Author	Title	Year	Findings	Relevance to Project
1	ZHAOSHENG ZHU ; GUOHAN LU ; YAN CHEN ; ZHI JUDY FU ; PHIL ROBERTS ; KEESOOK HAN , , 2008, 32nd Annual IEEE International Computer Software and Applications Conference: 425-436	Botnet Research Survey	2008	The paper presents a survey of recent advances in botnet research	Helpful in developing a general idea of internet bots.
2	Andreas Schaefer, Marc Reichenbach Dietmar Fey, 2012, In: Proceedings of IEEE Visualization, pp 190–198	Continuous Integration and Automation for Devops	2016	This work presents insights in how automation can be used to improve scalability and testability while simultaneously reducing the operators' work.	Provides a general idea of Development Operations and its need in the software development cycle.

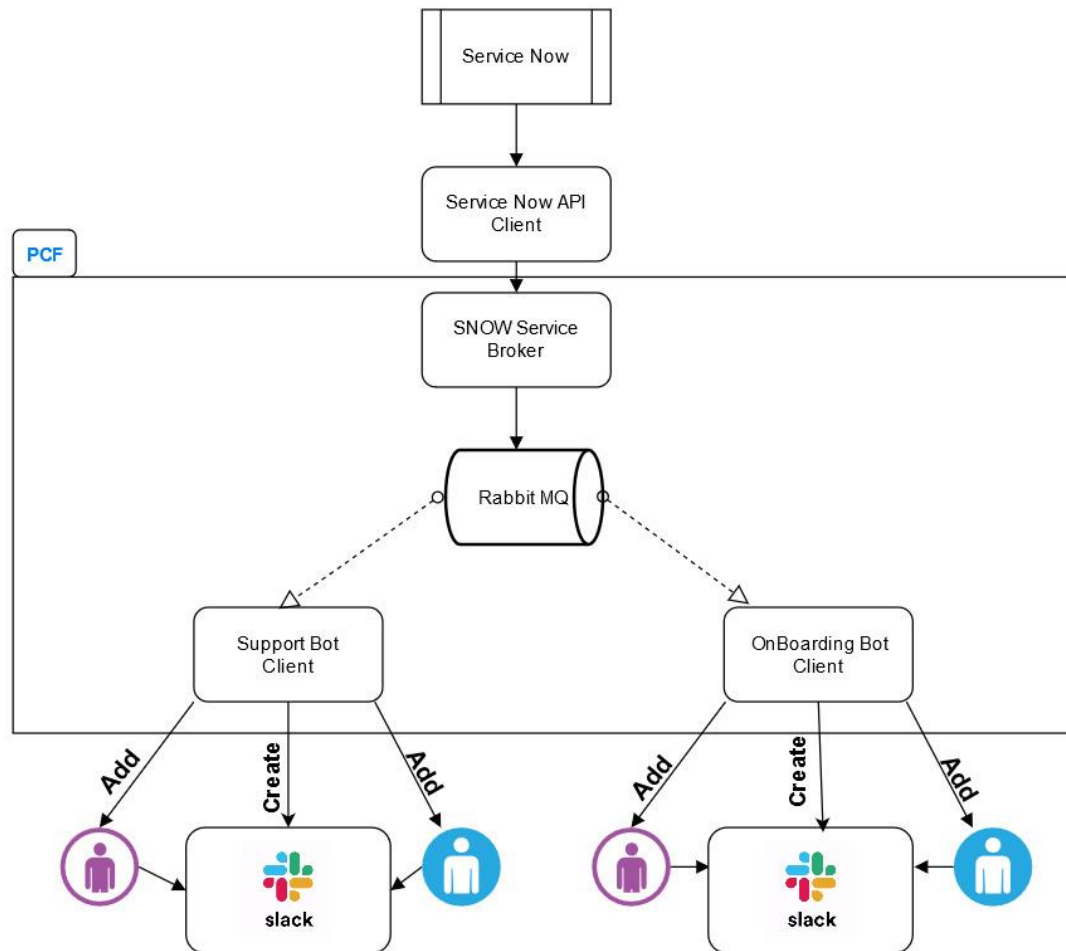
Problem Definition



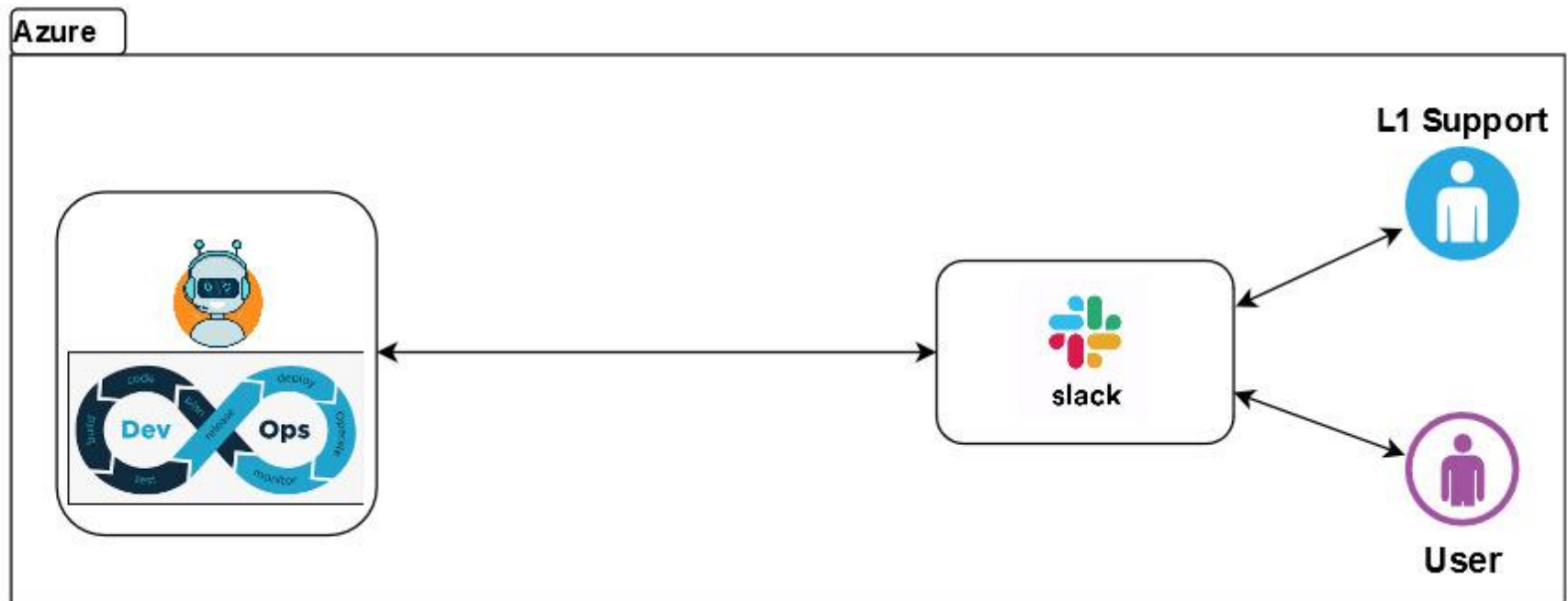
Problem Definition

- Customer will raise the ticket on Service Now
- Service Function will fetch the ticket and will undertake ticket segregation using sorter mechanism
- Depending upon type of ticket, appropriate action will be taken.
- The bot will create a Slack channel and will provide a questionnaire to the end user
- Depending upon the answers, concerned L1 support person will be added to the channel.

Proposed Solution



Proposed Solution



Proposed Solution

- Once the ticket has been raised on the Service Now Page, the Service Now API Client will fetch those tickets
- It will then hand them over to the SNOW Service Broker
- The broker is responsible for feeding them into a Rabbit Message Queue.
- The Sorter works on this queue to segregate the tickets and depending upon the type, the ticket is passed onto the Bot client.
- There are different bot clients for different ticket types

Proposed Solution

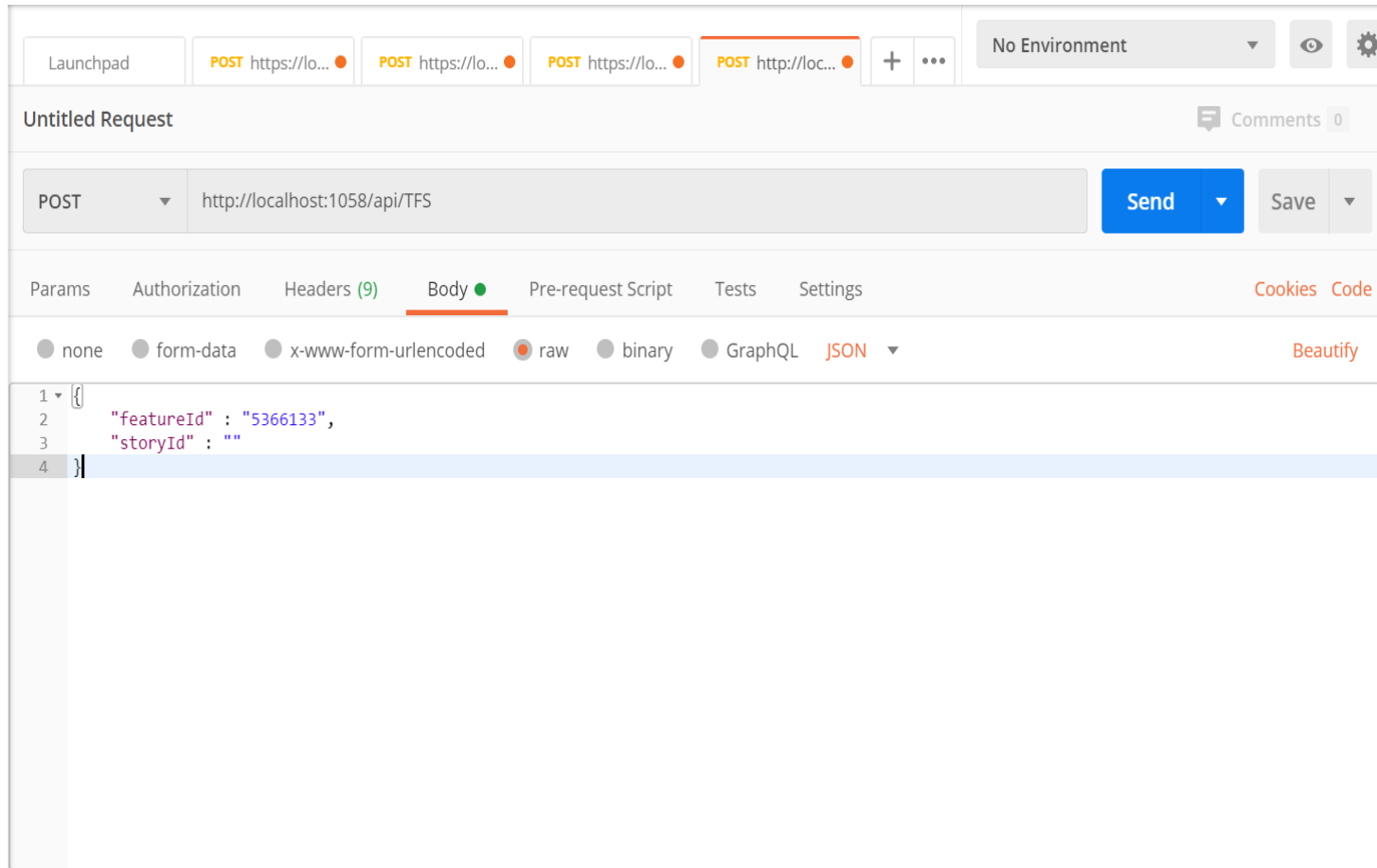
- The client will then create a Slack Channel and add the customer and the L1 support person to it.
- The Bot client and the Service Broker have been deployed in Pivotal Cloud Foundry (PCF)
- On the other hand, the created channel has been deployed in Azure.
- Azure provides certain add-on features which makes the process of using Slack API for interacting with Slack, easier.

Tech Stack Overview

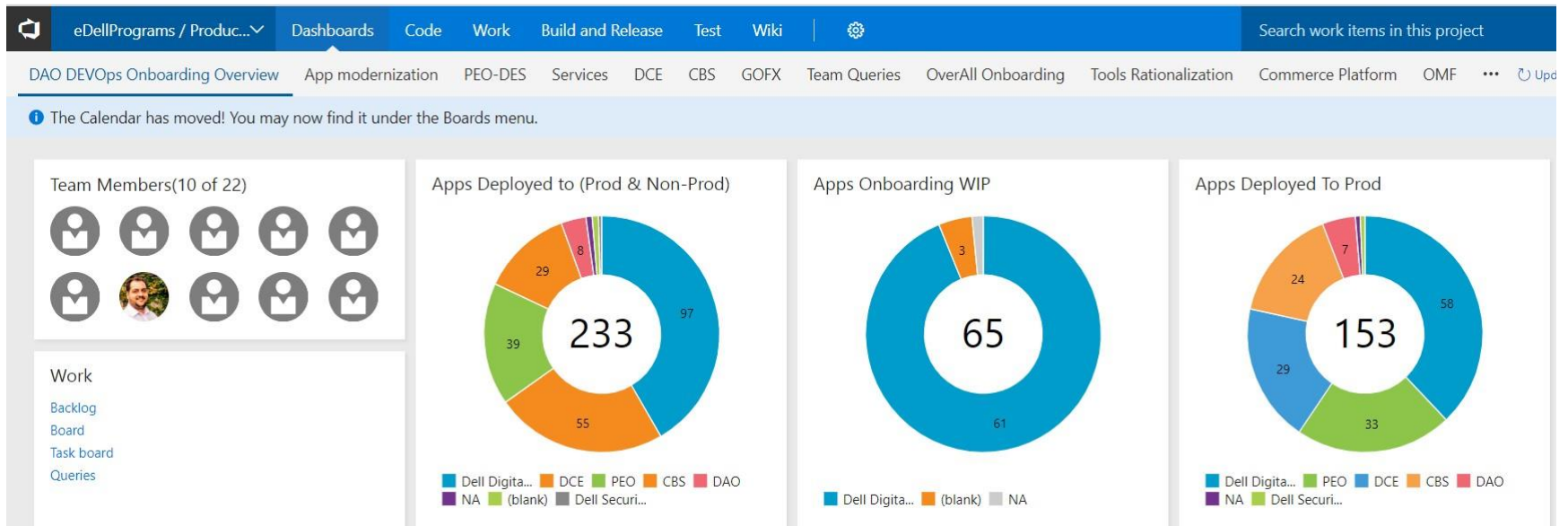
Visual Studio

- The code has been written in Visual Studio IDE.
- It provides the Bot Framework SDK which contains templates which are helpful in the development process.
- Since the project developed is being deployed to the back-end of the concerned application, the programming language used is C Sharp(C#)

Postman



Team Foundation Server



Team Foundation Server

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Add link

You are adding a link from:

   8109864 Onboarding FIL- [Traditi...
Updated 4/28/2020,  Cancelled

Link type

Affected By

☒ Work item type

Task

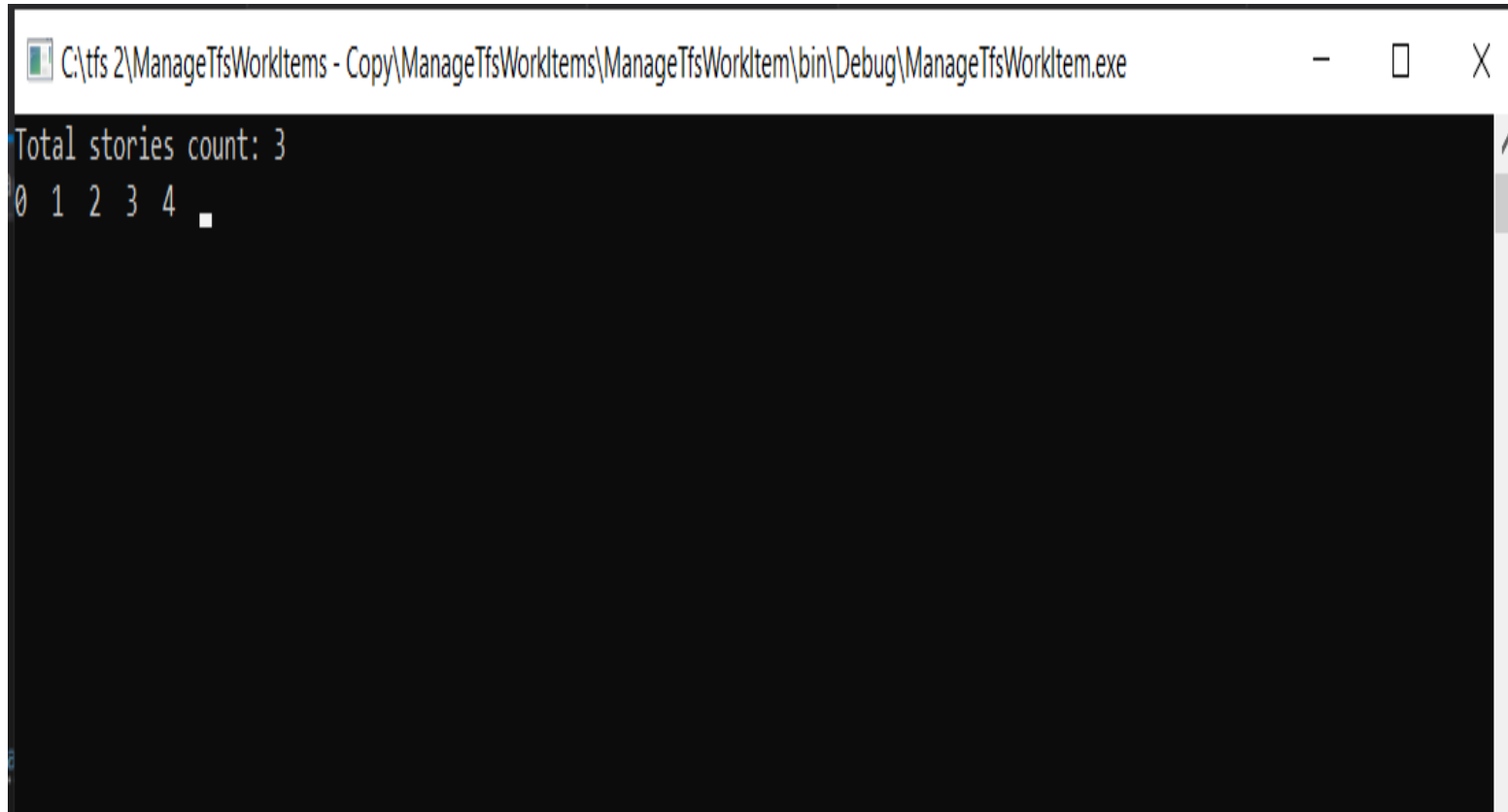
Title



OK

Cancel

Team Foundation Server



A screenshot of a Windows command prompt window. The title bar shows the file path: C:\tfs 2\ManageTfsWorkItems - Copy\ManageTfsWorkItems\ManageTfsWorkItem\bin\Debug\ManageTfsWorkItem.exe. The command prompt displays the text "Total stories count: 3" on the first line and "0 1 2 3 4 ." on the second line, with a cursor at the end of the second line.

```
C:\tfs 2\ManageTfsWorkItems - Copy\ManageTfsWorkItems\ManageTfsWorkItem\bin\Debug\ManageTfsWorkItem.exe
Total stories count: 3
0 1 2 3 4 .
```

Slack



Hello [redacted]

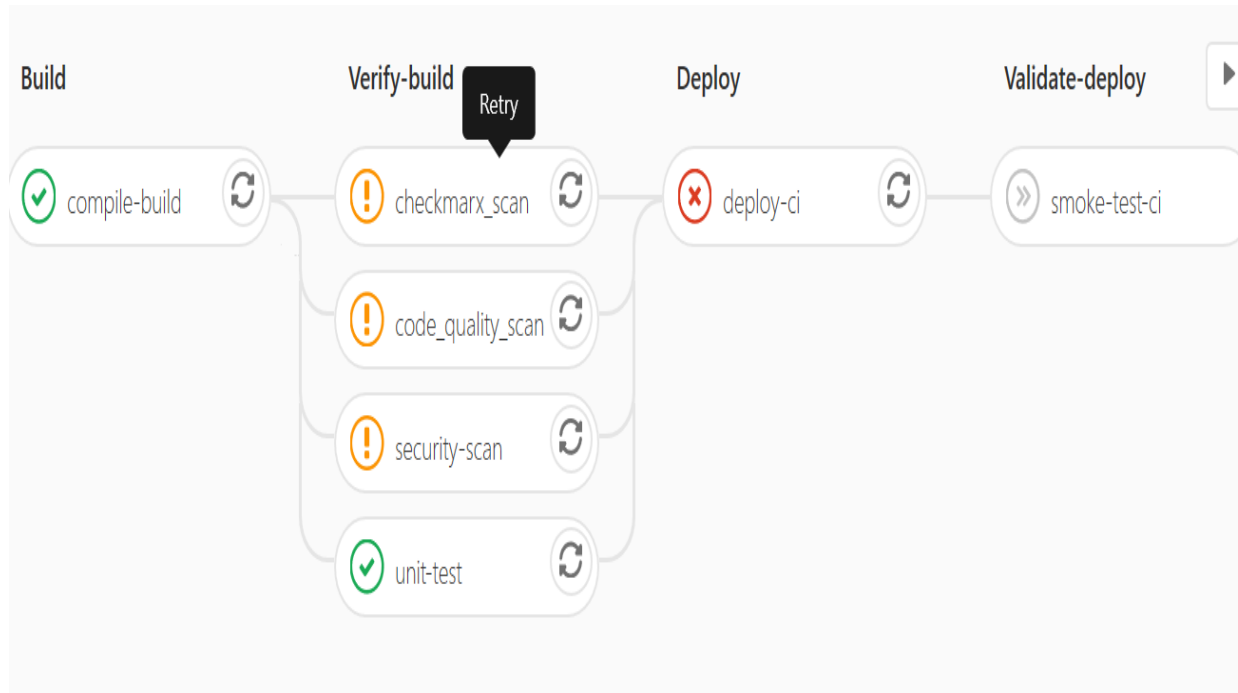
This channel is dedicated to your Service Now Request [228 \(GitLab Support Request\)](#).

Click on **Start** button below to input the detail required for processing the support request. When you are satisfied with the support engagement, click on **Archive** button to decommission the channel and close the Service Now request.

All conversation in this channel will be logged to the Service Now request for reference. This message has been pinned to the channel, click the pin icon in the Slack channel at any time to view this message again.

Start **Archive**

Gitlab



Deployment

- Continuous Integration and Continuous Deployment (CI/CD) is carried out using Gitlab.
- Different team members work on the project locally and push their changes to the project.
- Once the pipeline tests have been successfully passed, the project is deployed to the environment where various teams can use it commercially.
- This phase of the project is still under progress.

Conclusion

- DevOps is bridging the gap between developers' need for change and operations' resist to change
- The primary purpose of the project is reducing human involvement in the tiring work of support assistance.
- If the bot developed through this project works in the manner intended, it can serve as a major milestone in the company's journey towards automation.
- The resources used in the development of this project have a very high value in terms of future implementation.

Future Scope

- The migration of channel creation from Slack to Microsoft Teams which is gradually becoming the choice of interaction in the organisation.
- The development of knowledge base in Microsoft Graph API to achieve migration to Microsoft Teams.
- All the tools supported by DevOps team have not been considered for this project's development. The addition of non-compatible tools to the bot's servicing domain is expected in future.
- The innovations in Chatbot technology will bring a greater need to improve upon the product which has been developed.

Acknowledgment

I would like to express my sincere gratitude and appreciation to the Electronics and Communication Department, SMIT, for providing me with the opportunity of undertaking a major project at Dell Technologies and to our HOD Dr. Saurab Dhar, for his support.

I would like to mention my special thanks to my project guide, Arun Kumar Singh, for his meaningful contribution to this work.

I would also like to thank my manager Mrs. Sailaja Karri and Mr. Subrahmanyam Kirikera who acted as my mentor throughout the internship and was a constant support.

Finally, I would like to extend my gratitude to all my batchmates and friends at Sikkim Manipal Institute of Technology, for providing their support in various manners during my internship.

References

[1][Zhaosheng Zhu](#) ; [Guohan Lu](#) ; [Yan Chen](#) ; [Zhi Judy Fu](#) ; [Phil Roberts](#) ; [Keesook Han](#).

"Botnet Research Survey." *2008 32nd Annual IEEE International Computer Software and Applications Conference*: 425-436.

[2] Childs H, Brugger ES, Bonnell KS, Meredith JS, Miller M, Whitlock BJ, Max N (2005) A contract-based system for large data visualization.*In: Proceedings of IEEE Visualization, pp 190–198*

[3]Delaet T, Joosen W (2007) Podim: a language for high-level configuration management.*In: Proceedings of the 21st conference on Large Installation System Administration Conference, Berkeley, CA, USA, USENIX Association, pp 21:1–21:13*

Thank you

Dear Harsh Agarwal,

This is to confirm that you have been interning with Dell Technologies.

Your Internship details are mention below :

Project Name	Support BOT for Automating L1 Triage
Project Start Date	January 20, 2020
Project End Date	May 29, 2020
Location	Hyderabad
Reporting Manager	Karri Sailaja
Guide Name	Subramanyam Kirikera
Sponsor Name	Cody Taylor

You have successfully completed the project work with Dell Technologies. Your findings in course of the project has been found to be practical and relevant. Some of the recommendations will be incorporated on approval from the business.

Your performance during the tenure of the internship has been found to be satisfactory.

Note: You can use this email as a confirmation/certificate of internship completion in case the college requires.

Best Regards,
Dell University Relations – India
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